Meiqi Zhang

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Summary

Financial Mathematics & Statistics graduate from the University of Sydney (GPA 3.7, Dalyell Scholar) with hands-on experience in large-scale mathematical optimization, stochastic programming, and quantitative finance. Implemented cardinality-constrained portfolio MIQP with Lagrangian relaxation, two-stage stochastic programs solved via the L-shaped (Benders) method and progressive hedging, and LLM-guided reinforcement learning for portfolio management. Research output spans EI-indexed first-author publication and submissions to EMNLP and CHI.

Education

Nanyang Technological University — Singapore
M.Sc., Financial Technology — *Aug. 2026*

The University of Sydney — Sydney, Australia
B.S., Financial Mathematics and Statistics — GPA: 3.7/4.0, Dalyell Scholar — *Feb. 2025 – Jun. 2026*

Shanghai Jiao Tong University — Shanghai, China
Exchange Program, Statistics — Mathematical Finance, Stochastic Processes — *Dec. 2025 – Feb. 2026*

Optimization Projects

Large-Scale Portfolio Optimization: Cardinality-Constrained MIQP & Lagrangian Relaxation — Feb. 2026 – Apr. 2026
Python, Gurobi, CVXPY/OSQP, PuLP | GitHub

- Extended classical Markowitz mean–variance optimization to the NP-hard cardinality-constrained setting (hold at most K assets): formulated the MIQP with big-M linking constraints and solved it exactly with Gurobi (PuLP/CBC fallback) across K values on S&P 500 constituent data.
- Designed a Lagrangian decomposition relaxing the linking constraints $x_i \leq z_i$: the x-subproblem reduces to a convex QP (CVXPY/OSQP) and the z-subproblem to a closed-form greedy selection; updated multipliers via Polyak subgradient steps, achieving $< 2\%$ optimality gap at a fraction of MIQP solve time.
- Ran a scalability study ($N = 50 \to 500$ assets) showing Lagrangian relaxation scales where branch-and-bound becomes intractable; validated with a rolling out-of-sample backtest (return, volatility, Sharpe, turnover), Ledoit–Wolf covariance shrinkage, and a robust-optimization extension with ellipsoidal uncertainty sets.

Two-Stage Stochastic Portfolio Optimization via the L-Shaped Method — Apr. 2026 – Jun. 2026
Python, CVXPY, SciPy | GitHub

- Formulated portfolio choice under return uncertainty as a two-stage stochastic program (SAA over $S = 500$ bootstrap scenarios from real ETF data): stage-1 here-and-now allocation on the simplex, stage-2 recourse for shortfall and transaction costs, with CVaR_α risk linearized per Rockafellar–Uryasev.
- Implemented the L-shaped method (Benders decomposition for stochastic LPs): master problem accumulates optimality cuts built from stage-2 LP duals; tracked upper/lower-bound trajectories to certify convergence, in both single-cut and multi-cut variants.
- Implemented progressive hedging as a scenario-decomposition alternative; quantified the Value of the Stochastic Solution (VSS) and Expected Value of Perfect Information (EVPI), with an out-of-sample rolling backtest showing the SP portfolio dominating the expected-value portfolio on CVaR and Sharpe.

Research Experience

GIFT: LLM-Guided State–Reward Interface Design for Financial Reinforcement Learning — Mar. 2026 – May 2026
Core member | submitted to ACL ARR May 2026 (target: EMNLP 2026)

- Proposed the GIFT framework embedding LLM code generation into a PPO portfolio-RL pipeline; the LLM designs state and reward interfaces rather than acting as the trading agent, avoiding online-inference latency and instability.

- Built factor-guided state augmentation, risk-rule reward shaping, and diagnostic-driven refinement modules; rolling-window evaluation showed improved out-of-sample risk-adjusted returns over multiple baselines.

Mixed-Initiative AI Assistant for Retail Investors Reading 10-K Filings May 2026 – Present
First author (in preparation) | target: CHI

- Derived a three-tier difficulty framework for retail 10-K reading from interview and diary studies; designed a dual-panel system integrating in-situ explanation, reasoning visualization, and source tracing.
- Three controlled experiments ($N = 24 \times 3$): integrated interface improved efficiency 20%; approval-based reasoning raised hallucination detection from 52% to 78% with trust-calibration error 0.18 (vs. 0.42 black-box).

Time-Series Analysis Software for Electrochemical Hydrogen Bubble Dynamics Feb. 2026 – Jun. 2026
Core developer

- Independently developed a Python analysis system (Savitzky–Golay denoising, Welch PSD estimation for non-stationary signals) locating bubble detachment frequencies; built a photoelectric synchronization module aligning current signals with optical imagery at frame level, cutting manual annotation by ∼80%.

AutoInteract: Cross-Cultural LLM Human–Computer Interaction via Meta-Interaction Steps Jul. 2025 – Aug. 2025
First author | published at IEEE CIPAE 2025 (EI-indexed)

- Proposed the AutoInteract framework decomposing complex cross-cultural tutoring tasks into sequences of meta-interaction steps; constructed the CLeIDS dataset, with task success rate exceeding ReAct and TPTU baselines by ∼47% at 69B scale.

SLAM Point-Cloud Segmentation and Vegetation Detection with Neural Networks Sep. 2024 – May 2025
Research assistant | second author, SCI JCR Q2 journal

- Fine-tuned PointNet/PointCNN architectures and loss functions and optimized vegetation point-cloud feature inputs, improving IoU by 12%.

Internship Experience

Daphne Capital Remote
Quantitative Intern *Jun. 2025 – Aug. 2025*

- Developed an LSTM-based Bitcoin price-prediction model covering data cleaning, feature engineering, and backtesting end-to-end; validated via cross-validation and sensitivity analysis, and stable out-of-sample directional accuracy led to the signal's inclusion in the team's quantitative decision inputs.

Ginkgo Global Capital Management Remote
Secondary-Market Investment Intern *Feb. 2025 – Jun. 2025*

- Tracked macro fundamentals and fund flows via Wind and authored precious-metals strategy reports; built supply–demand balance sheets and miner AISC valuation models; backtested gold–silver ratio mean-reversion, delivering independent pricing views and stress-test reports to the investment committee.

China International Capital Corporation (CICC) Shenyang, China
Industry Research Intern *Sep. 2024 – Feb. 2025*

- Independently built DCF and comparable-company valuation models with financial forecasting and sensitivity analysis using Bloomberg comparables; tracked global IPO/refinancing activity, covered HK/US internet-sector fundamentals, and co-authored institutional in-depth research reports.

Technical Skills

Programming: Python (NumPy, pandas, scikit-learn, PyTorch/TensorFlow, Stable-Baselines3, CVXPY, PuLP), R, SQL
Optimization: MIQP/MILP modeling, Lagrangian relaxation, Benders / L-shaped decomposition, progressive hedging, stochastic programming, CVaR optimization, robust optimization; solvers: Gurobi, OSQP, CBC, HiGHS
Finance: financial statement analysis, DCF and comparable-company valuation, factor research, backtesting; Wind and Bloomberg terminals
Languages & Tools: English (fluent), Mandarin (native), Russian (fluent); Git, LaTeX, Claude Code, GitHub Copilot; Oracle MySQL 8.0 OCP Database Administrator